Proceedings of the 4th International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS-2026)
IEEE Xplore Part Number: CFP26DN7-ART; ISBN: 979-8-3315-7009-5

AI-Powered Waste Detection and Sorting using Object Detection

1st A Vibhas Reddy
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23004@bl.students.amrita.edu

2nd Hima Sagar B
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23014@bl.students.amrita.edu

3rd S Pavan Kumar
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23051@bl.students.amrita.edu

4th B Saranya Devi
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
b_saranya@blr.amrita.edu

2026 4th International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS) | 979-8-3315-7009-5/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICSSAS68835.2026.11559513

***Abstract*— The increasing rate in the plastic and municipal waste has posed a pressing pressure on the necessity of scalable and precise automated recycling systems. Current object detection-based waste sorting solutions are either computation ally intensive or cannot have reliable performance in real world heterogeneous settings. Despite growing research, limited comparative studies examine the trade-off between lightweight, hybrid, and transformer-based architectures for practical waste segregation. This work will attempt to design and test an AI based waste detection and sorting system by analytically com paring several deep learning architectures, such as lightweight YOLOv8 classifiers, YOLOv11 hybrid models, using ResNet and Swin Transformer, and attention-enhanced models with CBAM. A large scale of experiment is carried out on an annotated waste dataset by the evaluation metrics such as precision, recall, F1 score and computational cost. It was found that the YOLOv11 + ResNet hybrid has a high overall performance (F1 = 0.970) and the highest F1-score (0.9738) belongs to the YOLOv11 + Swin Transformer, which proves that transformer-based feature extraction is quite effective. Lightweight YOLOv8 models are still competitive with much less computation requirements, and attention modules have variable benefits. The results indicate that hybrid and transformer design improve the robustness of detection and make automated sorting of waste scalable, which leads to more efficient recycling and less negative effect on the environment.**

Keywords*— *Waste Detection, Object Detection, YOLOv11, Swin Transformer, ResNet, Automated Waste Sorting

I. INTRODUCTION

Rapid growth of municipal solid waste as a result of urbanization, industrialization and evolving consumption trends has emerged as a global concern. The lack of proper segregation of waste does not only decrease the level of recycling, but also causes environmental pollution, use of more landfills, and health hazards to sanitation employees. Most real-world applications continue to manually sort or process wastes or use rule-based mechanical methods which are most of the time inefficient, prone to errors and not safe.

One of the main constraints of the current waste management systems is that it is impossible to precisely and reliably sort waste into useful categories in large scale. Mixed waste streams have a substantial negative impact on the quality of recycling and the cost of their processing and automated and intelligent waste segregation is not a luxury but an essential requirement. Although recent developments in artificial intelligence, especially the object detection system with deep learning have promising solutions [1], their implementation is still difficult because of the variability in the appearance of waste, lighting, and occlusion, as well as the complexity of the background [2].

Among the modern object detection frameworks, the YOLO (You Only Look Once) family of object detection models has seen massive adoptions for its capability to perform real time detection with high accuracy [5]. Lightweight versions like YOLOv8n-cls are especially appealing to be used in embedded and edge-based waste sorting systems where the computational feasibility is a significant constraint. However, although the light models can deliver faster inference, they may find it difficult to extract deep visual features contained in heterogeneous waste materials, especially with different light conditions, occlusion and background noises usually appearing in waste streams in the wild.

To overcome these challenges, hybrid and attention-based deep learning architectures have become promising solutions to these problems [3]. Hybrid models employing both the YOLO based detection and powerful feature extractors such as ResNet could be used to increase the representational capacity and remain reasonably computationally efficient. Furthermore, transformer-based backbones such as Swin Transformer, intro duce global contextual modeling capabilities, which is suitable for the recognition of visually similar or partially occluded waste objects. Attention mechanisms such as Convolutional Block Attention Module (CBAM) have also been studied to only put emphasis on informative features of the space and channel features but their effectiveness may vary with object texture and classes distribution.

Despite these advances, there is still no comprehensive comparative studies, which analyze the trade-off between lightweight classifiers, hybrid CNN-based classifiers and

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

architectures, which includes transformers and attention-based uplifts, and are dedicated to waste detection and sorting applications. Most of the works available focus on a specific model or dataset, so the research is limited regarding the generalization of the model, the precision/recall trade-off and the suitability for application in practical waste management systems.

In this paper, an AI-powered waste detection and sorting architecture is proposed and evaluated based on multiple object detection and classification architectures that are based on YOLO matrices. We thoroughly perform an experimental analysis on YOLOv8 and YOLOv11 models including lightweight classifiers, hybrid CNN Integral with ResNet, transformer based Swin backbones, CBAM enhanced variants. The models are evaluated based on standard performance metrics, such as precision, recall and F1-score, to ensure that the models accurately identify waste categories. The results show that hybrid and transformer-based models significantly perform better than baseline and attention-only models resulting in a high F1 scores with robustness of waste types. The results of this study offer important insights for model selection and intelligent waste sorting systems, as well as move toward developing scalable, automated and sustainable waste management solutions.

Despite significant progress in deep learning-based waste classification, existing approaches face several practical limitations. Many studies focus on single-model architectures without systematically analyzing the trade-offs between lightweight models, hybrid CNN-based models, and transformer-based architectures. Furthermore, limited attention has been given to evaluating these models under consistent experimental conditions, making it difficult to identify the most suitable approach for real-world deployment.

Additionally, most existing works emphasize multi-class classification accuracy without aligning the problem formulation with practical waste management requirements, where the primary need is efficient segregation into biodegradable and non-biodegradable categories. This disconnect between research objectives and real-world applicability highlights the need for a more context-driven and comparative study.

To address the specified research gaps, the present study provides a comprehensive comparative analysis of AI-powered waste detection and sorting in the framework of the systematic assessment of lightweight, hybrid, attention-enhanced, and transformer-based YOLO models under a common experimental configuration. Unlike the previous works that focused on the evaluation of individual or few models, the current work presents the evaluation of the trade-offs between the efficiency of the model and the effectiveness of the waste detection.

Contributions of the present work: The present work will make the following contributions to the development of the AI-based waste management system:

This work proposes a deep learning-based approach to detecting and sorting waste between biological and non-biological waste in line with the requirements of realistic recycling rather than detailed classification of the material as in conventional methods.

Various architectures were compared systematically, such as YOLOv8n-cls with CBAM and Hybrid ResNet, YOLOv11, and YOLOv11 in conjunction with ResNet and Swin Transformer blocks.

It is concluded that YOLOv11 + ResNet is the best combination to achieve good results, although YOLOv8n cls can still be considered a reliable choice for low resource.

II. LITERATURE SURVEY

Rad et al. [1] stated that convolutional neural networks can effectively classify waste materials into recyclable categories, demonstrating that deep feature representations significantly outperform traditional handcrafted features for trash image classification tasks.

Yang et al. [2] stated that automated waste classification using deep learning improves sorting accuracy under real world conditions, particularly when trained on large-scale, diverse datasets containing variations in lighting and object orientation.

Zhang et al. [3] stated that transfer learning using pretrained CNN backbones enhances waste detection performance while reducing training time, making it suitable for deployment in practical recycling systems.

Ahmed et al. [4] investigated recyclable product classification using CNN and transfer learning models including DenseNet169, MobileNetV2, and ResNet50V2. The results demonstrated that transfer learning significantly improves classification accuracy, with ResNet50V2 achieving 98.95% accuracy, emphasizing the potential of deep learning for automated waste management systems.

Wang et al. [5] stated that YOLO-based object detection models achieve a strong balance between detection speed and accuracy, making them particularly effective for real-time environmental monitoring and waste management applications.

Li et al. [6] introduced an adaptive attention mechanism integrated with YOLOv3 and MobileNetv2 to improve object detection performance. Their approach achieved notable mAP improvements on KITTI and Pascal VOC datasets, highlighting the effectiveness of attention modules in enhancing detection accuracy.

Liu et al. [7] stated that transformer-based vision models capture global contextual information more effectively than conventional CNNs, leading to improved classification accuracy for visually ambiguous objects.

Mishra et al. [8] developed a hybrid DenseNet-121 + SVM model for smart waste classification using the TrashNet dataset. Their approach achieved 99.84% accuracy, highlighting the effectiveness of deep feature extraction combined with SVM for automated waste management systems.

Huang et al. [9] developed a hybrid CNN–Transformer network that integrates local and global feature fusion for object detection. Their model achieved strong mAP results on multiple remote sensing datasets, demonstrating the advantage of combining convolutional and transformer-based architectures.

Sivakumar et al. [10] proposed an IoT-based smart solid waste management system integrated with computer vision and machine learning techniques. Their approach employed a CNN and a transfer learning-based ResNetV2 model to classify waste into biodegradable and non-biodegradable

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

categories. The system incorporated batch normalization and hybrid pooling techniques to enhance stability and performance.

The recent developments in the field in 2024 and 2025 have witnessed a paradigm shift in the application of traditional CNN-based sorting approaches to hybrid and transformer-based approaches to address the requirements of Industry 5.0 in waste management. While the earlier approaches based on YOLOv3 and other ResNet variants demonstrated the viability of the application of deep learning in waste sorting, the latest research in the field has demonstrated the viability and superiority of the Swin Transformer in providing the necessary global reasoning capabilities to effectively classify visually similar recyclables, while the YOLOv11 framework has been demonstrated to provide the necessary speed to address the requirements in the field. However, the absence of a comparative study on the viability of the application of the recent approaches in the field has been a notable gap in the field, which the authors address in the paper by justifying the application of the hybrid YOLOv11 framework.

III. METHODOLOGY

The proposed methodology utilizes a multi-staged deep learning methodology that is optimized with the goal of striking the right balance between local spatial feature extraction and global reasoning in waste segregation. Unlike monolithic structures commonly found in deep learning methodologies, the proposed approach utilizes YOLO-based detection heads along with specific backbones such as ResNet-50 with residual learning capabilities and Swin Transformers with hierarchical attention-based feature fusion capabilities. The overall methodology is structured in the following way: (i) stochastic data augmentation with invariance in the class domain, (ii) transfer learning-based feature encoding, and (iii) binary classification with the help of a decoupled detection head in the domain of biological and non-biological waste segregation.

A. Dataset Description

The experiments were performed using the Garbage Dataset which is a publicly available dataset of images that were generated in order to perform waste classification experiments. The dataset is a total of 5,886 images and is divided into six types of waste, i.e., battery, biological, metal, paper, plastic and trash. Each class has around 939 to 983 images resulting in a relatively even distribution of data. The images are realistic models of real-world waste situations and contain large differences with respect to the background, illumination, orientation, and scale of objects. In addition, the dataset contains high inter-class similarity, for example, of paper and trash, and high intra-class variation, for example, of different forms of plastic waste, which makes the classification task particularly challenging.

Although the dataset contains six classes of fine-grained waste, the ultimate aim of this work has the potential to be binary waste segregation. For the purpose, the biological class has been considered under biological waste, and the remaining classes, such as battery, metal, paper, plastic, and trash under non-biological waste. This formulation fits the experimental configuration to the practical recycling needs, where the basic difference between biodegradable and non-biodegradable materials is most important.

B. Data Splitting Strategy

The data was categorized into training, validation and test data in the proportion of 70, 15 and 15 respectively for 4121 images. The stratification was done to maintain class distribution within all subsets.

Model parameters were learned using the training set, and hyperparameter tuning, model selection and convergence monitoring during training were done with the validation set. Validation loss was used to stop early in order to avoid overfitting. The test set remained fully invisible throughout the training and validation stages and was not used in the training and validation to guarantee no bias in the evaluation of the generalization of the model.

C. Image Preprocessing and Data Augmentation

All the images were resized to a fixed size to ensure compatibility with the deep learning architecture chosen, and also to maintain uniform input dimensions (224 × 224 pixels). Prior to the training process, pixel values were normalized in order to stabilize the gradient updates and speed up the convergence process. In order to overcome overfitting and to improve the generalization ability of the models, a set of data augmentation processes were implemented but only for the training set. These augmentations were random horizontal flips, random rotations, color jittering, random zoom operations, MixUp, and CutMix strategies. By making controlled changes in appearance and composition, these transformations allow the models to discover robust and invariant representations of features that are resilient to real-life noise and variability.

D. Model Architectures

As depicted in Figure 1, a series of deep learning algorithms are considered and analyzed based on transfer learning, where the initial parameters in the network are initialized through pre-trained weights acquired from large image datasets. This method allows the utilization of high-quality features with minimized computation effort. The architectures chosen are designed to combine both performance and efficiency, which make them appropriate for real-world use cases, such as edge-based sorting of waste items.

Lightweight CNNs such as YOLOv8n-cls are included in the proposed architecture due to the low computation effort required during inference. In addition to lightweight architectures, a series of more sophisticated algorithms based on YOLOv11 will be utilized. Hybrid architectures with YOLO heads coupled with sophisticated backbones (ResNet, Swin Transformer) are also considered in the research since convolutional architectures are more efficient at learning local features while transformer-based algorithms are able to detect global spatial relationships, which is important for separating waste categories that share visual similarities. The models will also include attention-based modifications using Convolutional Block Attention Module (CBAM).

The analysis focuses on comparing several YOLO-based architectures, including their modifications through changing backbones used in the models. All the experiments will be conducted under the same conditions (Figure 1).

The evaluated models are described as follows:

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

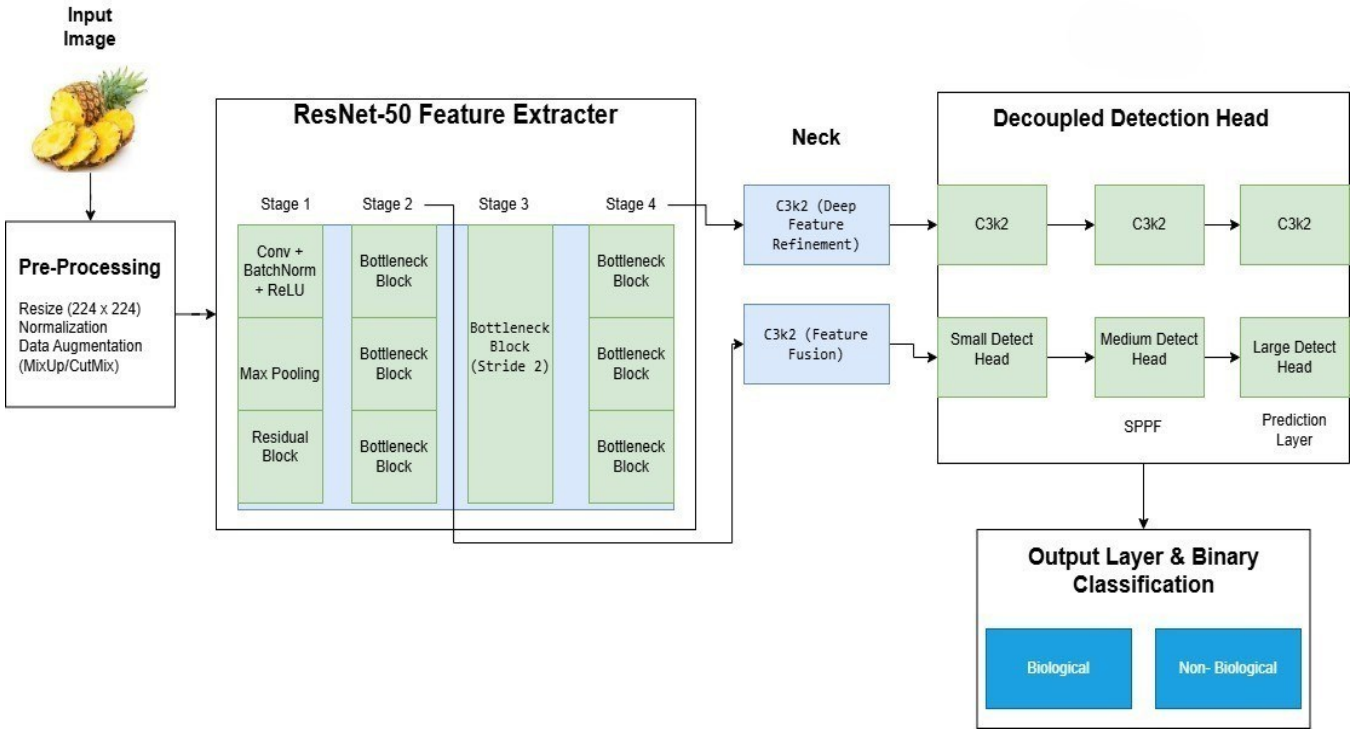


Fig 1. Overall model architecture of the proposed YOLO-based hybrid ResNet waste detection model.

YOLOv8n-cls (baseline):

It is a lightweight variant of the YOLOv8 algorithm designed to operate with minimal computational cost and inference speed. It will be used as the base model to test the efficiency of other methods on the dataset.

YOLOv8n-cls + CBAM:

In this model, a convolutional block attention module will be added to the baseline model to refine the extracted feature maps using channel and spatial attentions.

YOLOv8n-cls + Hybrid ResNet:

It is an improved baseline model where the backbone is changed from the lightweight one to ResNet based architecture for extracting hierarchical representations.

YOLOv11 (baseline):

A model based on the YOLOv11 is implemented as a stronger baseline compared to YOLOv8n-cls. The advantage of the model is its superior capability in detecting objects and extracting features compared to the previous baseline.

YOLOv11 + ResNet:

The backbone architecture is changed from YOLOv11 baseline model to ResNet to improve the learning capability of local spatial features, which increases the robustness of the model when dealing with occlusions and other adverse conditions.

YOLOv11 + Swin Transformer:

Swin Transformer model is employed as the backbone architecture in this variant to allow the network to extract information using long-range dependencies.

E. Training Procedure

Each model has been trained using supervised learning with binary cross entropy, since the problem was modeled as a binary classification problem.

$$L = -[\, y \, log(\hat{y}) + (1 - y) \, log(1 - \hat{y}) \,]$$

where y denotes the ground truth label and $\hat{y}$ represents the predicted probability of the positive class.

Optimization was performed using adaptive gradient-based optimization techniques to ensure stable and efficient learning. The training process was monitored using validation performance, and early stopping was employed to prevent overfitting. The learning rate was lowered step by step as training advanced - the steady drop helped the network settle and stay stable.

During testing, every item received one of two labels - biological or non-biological. The models had to place each piece of waste in the correct group plus had to remain reliable when faced with many kinds of waste.

F. Evaluation Metrics

To assess the effectiveness of the proposed models, accuracy, precision, recall, and F1-score were used to determine the performance of the models in an all-encompassing manner.

Accuracy is used to measure the general percentage of samples that are correctly classified. But in problems of waste classification where misclassification costs and distributions of classes can be widely differing, not only accuracy is required. Thus, accuracy and recall were employed to assess the accuracy and completeness of predictions respectively.

The F1-score which is defined as the harmonic mean of precision and recall was used as the main measure of comparing the models since it is a balance between false

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

positives and false negatives. This is mostly needed in the case of waste separation where misclassification would result in contamination of what is to be recycled.

All these metrics will present a powerful evaluation frame- work of comparing a variety of architectures to work under real-life circumstances.

IV. Results and Discussion

In this section, the evaluation of the different proposed models related to waste classification will be carried out. The evaluation will be conducted by using the metrics of accuracy, precision, recall, and F1 score. The results of the experiments clearly justify the efficiency of the hybrid models as well as the transformer models in the identification of waste, particularly binary classification of biological waste against nonbiological waste.

A. Performance Comparison Based on Accuracy

Accuracy represents the proportion of correctly classified samples among the total number of samples.

The classification performance of the models under test is illustrated through the use of training, validation, and test accuracy. In addition to the models under evaluation, the YOLOv11 model on its own was able to achieve test accuracy of 0.92.

Among all the models tested, the one with the best generalization performance would be the YOLOv11 + ResNet model, with the best training, validation, and test accuracy values of 0.9684, 0.9692, and 0.9741 respectively. With minimal differences between the training and validation accuracies, the learning performance would be excellent with minimal risks of overfitting, making it ideal for use in waste classification in the future.

YOLOv11 + Swin Transformer [11] is yet another excellent performer with a test accuracy of 0.9734. While this model has the best training accuracy of 0.9812, its relatively lower test accuracy suggests a certain amount of overfitting or data dependence. Nevertheless, the achieved test accuracy proves that transformer models achieve outstanding results when it comes to managing global context, which is crucial for differentiating similar looking waste products based on their visual characteristics. Additionally, the hierarchical window-based attention scheme used by Swin Transformer allows the model to concentrate on relevant image parts while preserving spatial information, thus tackling the problem of scale differences, lighting, and orientation changes present in waste materials.

YOLOv8n-cls model performs quite decently with regards to test accuracy as it has a test accuracy value of 0.9606. This would mean that efficient models perform excellently in classification tasks. However, the test accuracy results of the YOLOv8n-cls + CBAM model are very poor, with an accuracy of 0.9189 compared to the base model of YOLOv8n-cls. This shows that there are some wastes which may not necessarily require attention mechanism.

The YOLOv8n-cls + Hybrid ResNet [12] is an improved baseline model based on the YOLOv8n-cls with a test accuracy score of 0.9620. It Fig.2 shows the efficiency of deep learning in feature extraction, but it has poor results when compared to hybrid YOLOv11-based models.

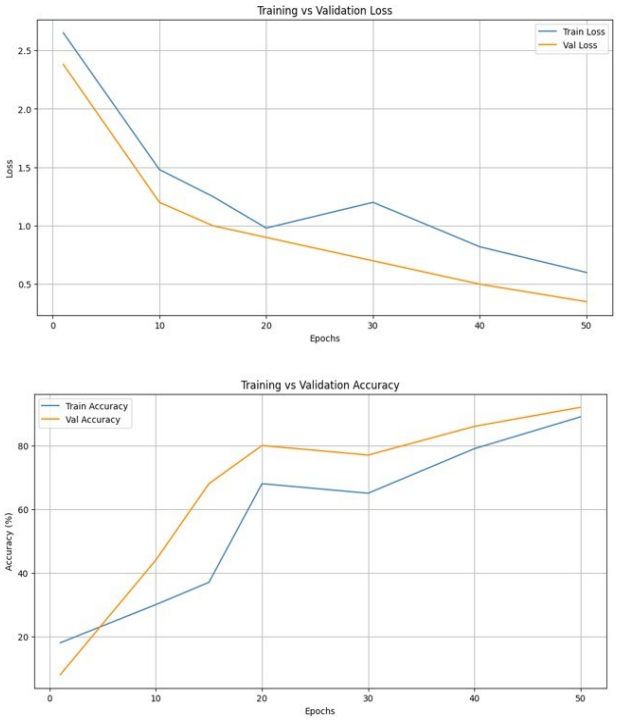


Fig. 2. YOLOv11 model

TABLE I
Accuracy Comparison of Waste Classification Models

Model	Test Acc
YOLOv8n-cls	0.9606
YOLOv8n-cls + CBAM	0.9189
YOLOv8n-cls + Hybrid ResNet	0.9620
YOLOv11 (Baseline)	0.9200
YOLOv11 + ResNet	**0.9741**
YOLOv11 + Swin Transformer	0.9734

In Table 1 comparison with other models, the stand-alone version of YOLOv11 shows lower results in test accuracy than hybrid ones but still has higher scores than attention-based YOLOv8 models. That can be explained by the fact that structural modifications of the model improve the feature representation capability of YOLOv11 even without using hybrid backbones.

The success of the YOLOv11 + ResNet model is caused by its possibility of hierarchically extracting spatial features thanks to deep layers. The use of residual blocks in ResNet architecture prevents the vanishing gradient problem, allowing the model to extract more discriminative features. Such characteristics are necessary for waste classification since objects of the same class differ significantly among each other. (Figure 3).

The YOLOv11 + Swin Transformer model shows higher scores in training than testing accuracy, which means that it is highly sensitive to dataset characteristics. That can be connected to the need of having a large dataset in order to train transformer-based models successfully without overfitting (Figure 4).

YOLOv11 + ResNet model outperforms YOLOv8n + CBAM model by around 5.5% in terms of accuracy, proving

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

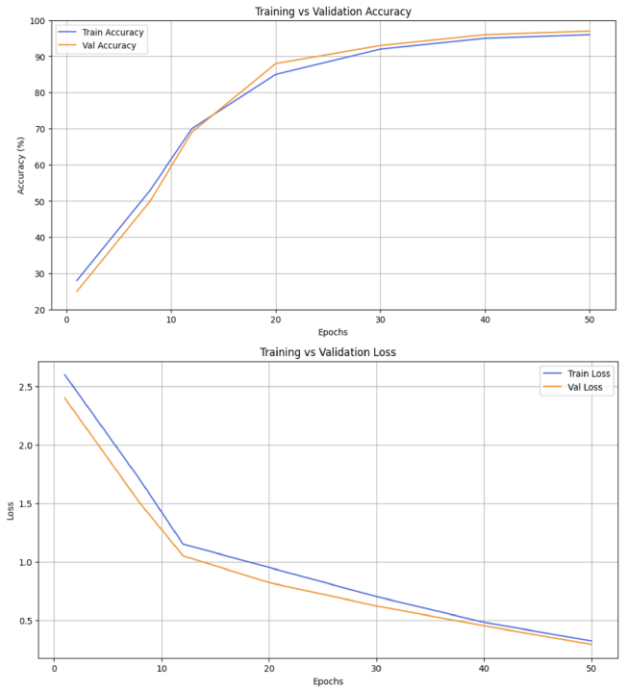


Fig. 3. YOLOv11 + ResNet model

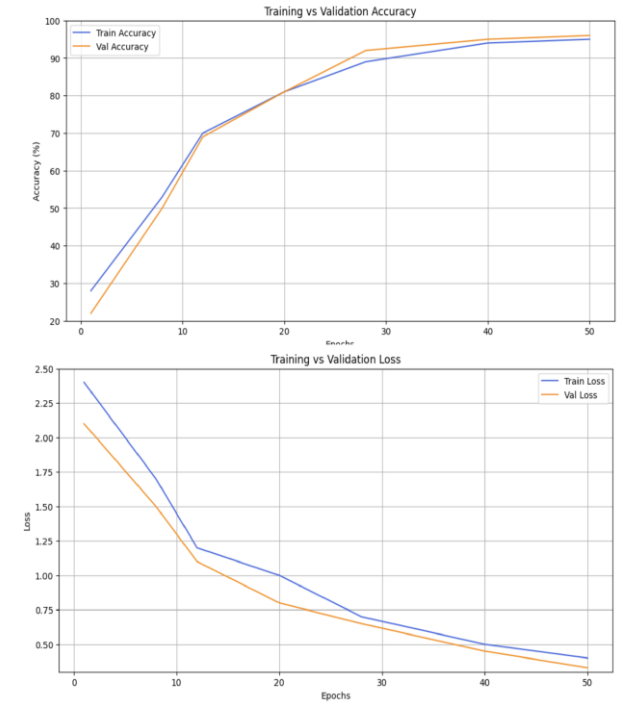


Fig. 4. YOLOv11 + Swin Transformer.

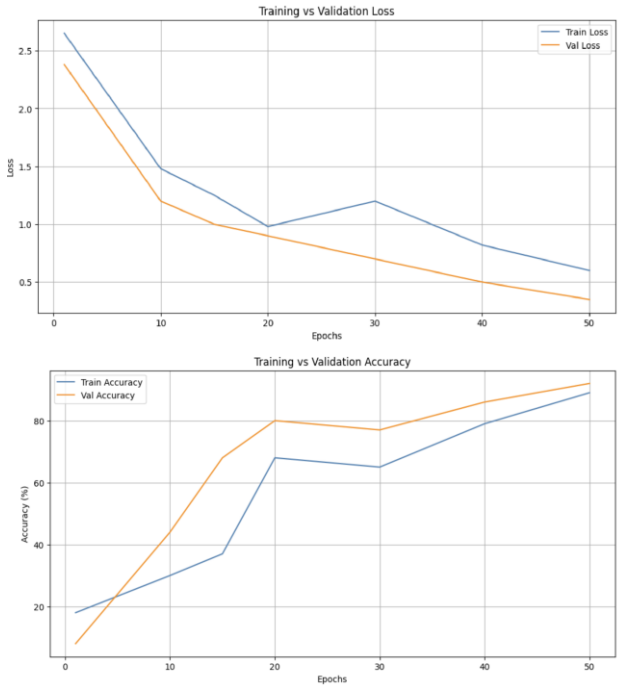


Fig. 5. YOLOv8n-cls model.

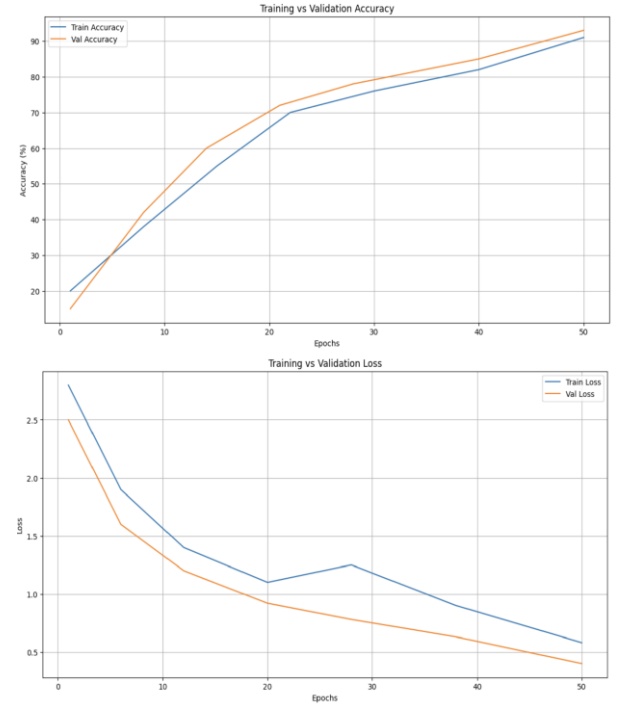


Fig. 6. YOLOv8n-cls with CBAM.

the efficiency of hybrid models over just attention-based improvements (Figure 5).

B. Precision, Recall, and F1-Score

Accuracy, precision, recall, and F1 score were analyzed aside from these metrics because these can give a more complete picture of model effectiveness, particularly since misclassification costs can be asymmetrically weighed during waste sorting. The YOLOv11 + ResNet combination model yielded a precision of 0.976, recall of 0.968, and F1 score of 0.970, which indicates a harmonious relation between correct sorting and the prevention of misclassifications, particularly false positives and false negatives.

Additionally, the YOLOv11 stand-alone model performs quite well with an average precision score of 0.93, recall score of 0.93, and F1-score of 0.93. Although it does not perform as well as YOLOv11 hybrid models, it still significantly beats the attention-based YOLOv8 models, suggesting that the architecture of YOLOv11 has some benefits over YOLOv8.

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

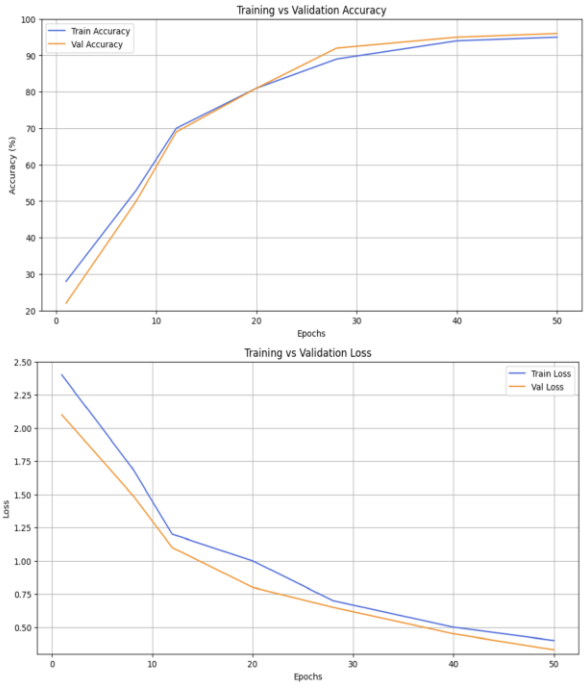


Fig. 7. YOLOv8n-cls with Hybrid ResNet.

Model YOLOv11+ Swin Transformer provides the highest value of the F1 measure at 0.9738, backed by a precision of 0.9791 and recall of 0.9685. In particular, its high recall value proves its efficiency in processing a large number of samples of waste data, and high precision indicates correct decision making in the class (Figure 6). This further supports the superiority of transformer models in understanding global dependencies, especially in visual scenes.

The precision, recall, and F1-score of YOLOv8n-cls are each 0.96, indicating a good performance of this model, keeping in mind that its architecture is quite simple compared to other models.

The precision, recall, and F1-score for YOLOv8n-cls + CBAM are lower with values of 0.92. Although the ability to concentrate on distinct features for certain classes can be enhanced by the utilization of attention modules, it appears that the amplification of noise features in heterogeneous visual categories like plastic and trash [13] might be happening through the application of CBAM [14].

The YOLOv8n-cls + Hybrid ResNet has a precision value of 0.9843, which expresses strong confidence in the output predictions. Although its recall measure of 0.9342 is lower comparatively, it means that some of the waste instances are missed, which signifies decreased efficiency regarding the elimination of all instances of waste (Figure 7).

This increase in precision in the YOLOv11 + Swin Trans former model shows the model's effectiveness in avoiding false positives by utilizing the contextual relationship between the visual features. This would be useful in distinguishing between visually similar categories of waste materials such as paper and plastic.

Although the model has slightly lower recall than its precision, this shows the model might be missing some data. This might be because the model uses global attention mechanisms that might not pick up on local features. The

TABLE II
PRECISION, RECALL, AND F1-SCORE COMPARISON

Model	*Precision*	*Recall*	*F1-Score*
YOLOv8n-cls	0.9600	0.9600	0.9600
YOLOv8n-cls + CBAM	0.9200	0.9200	0.9200
YOLOv8n-cls+ Hybrid ResNet	0.9843	0.9342	0.9641
YOLOv11 (Baseline)	0.935	0.93	0.93
YOLOv11 + ResNet	0.9760	0.9680	0.9700
YOLOv11 + Swin Transformer	0.9791	0.9685	0.9738

YOLOv11 + ResNet model has a more even balance between the precision and recall of the model. Table II shows the effectiveness of convolutional feature extraction in picking up both fine-grained and structural features in the heterogeneous waste materials.

C. Comparative Analysis with Baseline Methods

The proposed waste detection and sorting model architectures have been evaluated through a comparative study against several baseline models. The aim of this analysis is to determine the increase in accuracy achieved by both hybrid models and transformer models over lightweight models and attention-based models. The comparison of baseline models is also necessary to confirm the improvements made by the proposed models are not due to random fluctuations but due to the differences in architecture and features extraction capabilities.

In this work, YOLOv8n-cls is considered as a basic baseline model since it is a very light-weight model based on mini mum computation complexity and edge device applicability. In addition, the standard YOLOv11 is selected as a strong baseline model with enhanced feature extraction capabilities over the YOLOv8 model. Finally, YOLOv8n-cls with channel and-spatial attention mechanism (CBAM) is selected as an attention baseline model to determine the effect of attention mechanisms on classification robustness. YOLOv8n-cls with hybrid ResNet is selected as another baseline model to measure the effect of deeper convolutional layers in extracting features without using transformer-based global reasoning.

All mentioned above advanced architectures were tested under the same training conditions, using the same datasets, splitting, and augmentation strategies for a fair comparison. It can be seen from the table that all proposed hybrid models exhibit better metrics of accuracy, precision, recall, and F1 score compared to basic models and YOLOv11 architecture in general. YOLOv11 + ResNet model exhibits the best accuracy and recall, which means that classification is relatively stable and not affected by overfitting. In turn, YOLOv11 + Swin Transformer model has the highest F1-score due to its superior abilities in capturing global context dependencies between images.

Figure 8: Screenshot panels from the Waste Biodegradability Classifier web application. Rows (A) and (B) demonstrate real-time object classification results for diverse biodegradable and non-biodegradable waste items, respectively.

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

On the other hand, despite being much lighter and cheaper, YOLOv8n-cls model is also quite effective in classifying objects. However, it demonstrates poor results while processing images with mixed waste textures and occluded objects. CBAM-based version of YOLOv8n-cls was unable to achieve better results in contrast to the base one because attention mechanisms do not show high effectiveness in highly dynamic and heterogeneous environments. On the contrary, ResNet baseline works better than the lightweight baseline but per forms worse than both proposed hybrid models.

In conclusion, from the comparative study, it is evident that although lightweight models may be preferred in constrained environments, the single model YOLOv11 has proved its worth as a starting point in classification tasks, whereas the use of hybrid models based on CNNs and transformers improves classification accuracy and consistency. Overall, it can be said that the developed models outperform the conventional baselines used here in terms of performance improvements.

D. Discussion and Key Observations

The findings from the experiment provide several key technical insights. Firstly, the use of hybrid models based on YOLO and deep convolutional networks like ResNet is seen to provide an effective trade-off between features and generality. This is because both models have been able to effectively learn both low-level features like texture and high-level features like semantic information.

Secondly, transformer models have been seen to provide better performance in effectively dealing with global contextual relationships, which is important in distinguishing between visually ambiguous categories of waste. This is, however, seen to be dependent on the availability of sufficient data, as transformer models have been seen to be vulnerable to overfitting.

Thirdly, the use of attention mechanisms like CBAM is seen not to provide any improvements in performance. This is because of the high visual noise present in the data, where the use of attention mechanisms like CBAM could have led

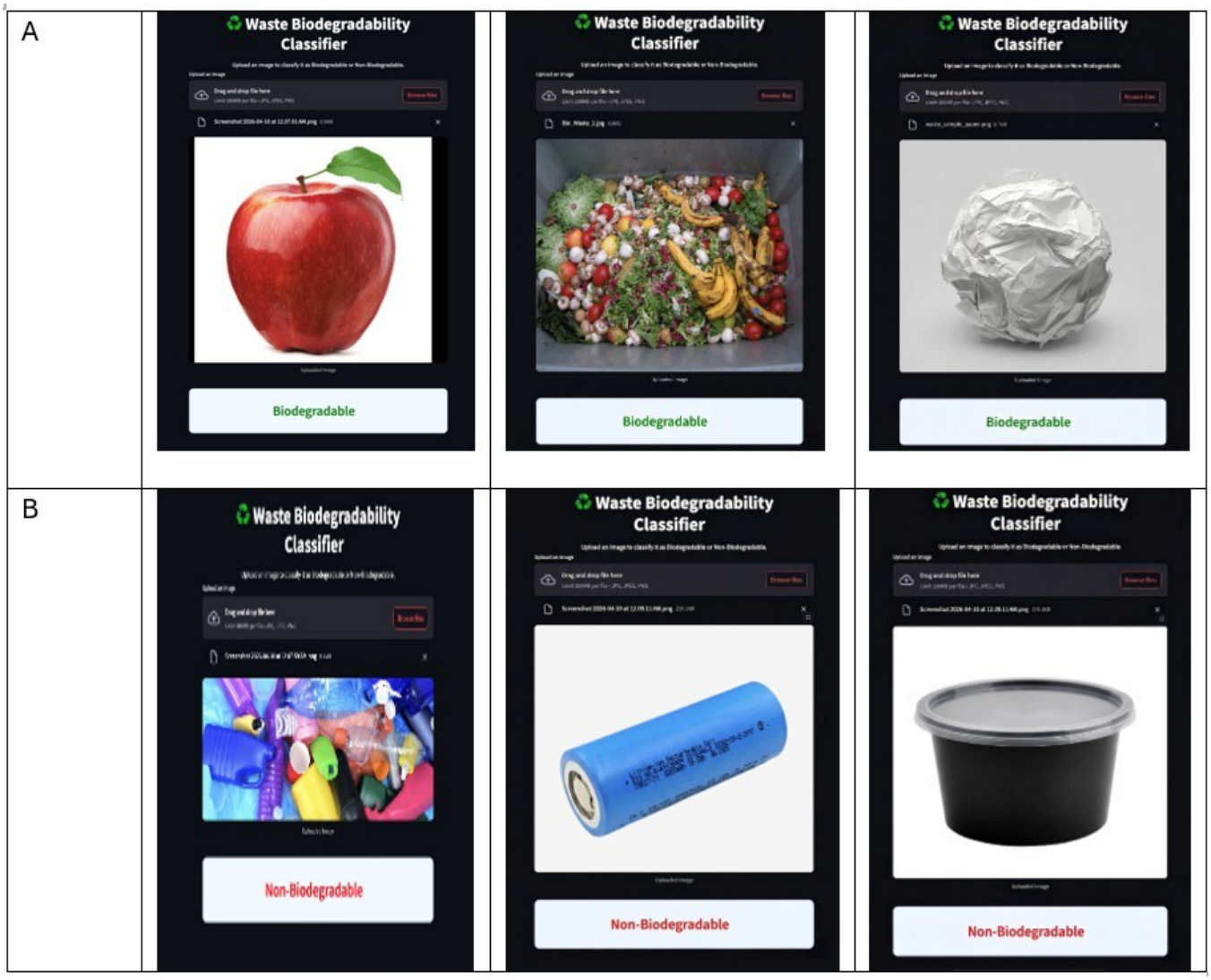


Fig. 8. A) Sample outputs for biodegradable waste classification using YOLOv11 + ResNet; (B) Sample outputs for non-biodegradable waste classification using YOLOv11 + ResNet.

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.

to the amplification of irrelevant features rather than their suppression.

Lastly, the use of lightweight models is seen to be highly relevant, as they provide an effective trade-off between computational efficiency and acceptable accuracy.

V. Conclusion

The research introduced an automated waste detector and segregation system based on AI through the use of transfer-learning-based YOLO architectures to process binary classification of the Garbage Dataset of biological and non-biological waste. A series of comparative experiments using lightweight, hybrid, attention, and transformer models showed that hybrid architectures performed better. The best test accuracy detected by the YOLOv11 + ResNet model was 97.41% with F1-score of 0.970, and indicates high generalization to the real world application. The transformer-based global features extraction was proven effective as the YOLOv11 + Swin Transformer had the highest F1-score of 0.9738. Light YOLOv8n-cls models retained comparable accuracy (96.06) at much less cost of computation, which is appropriate to edge settings, but CBAM attention variants did not do as well (92% accuracy). In general, all the findings assert that transformer-based and hybrid YOLO structures offer optimal trade-offs between accuracy and efficiency in scalable intelligent waste management. Further research will develop the system to use multi-camera real-time conveyor-belt implementation, and bigger datasets.

REFERENCES

[1] M. S. Rad, A. von Kaenel, A. Droux, F. Tieche, N. Ouerhani, H. K. Ekenel, and J.-P. Thiran, “A computer vision system to localize and classify wastes on the streets,” in Proc. Int. Conf. Computer Vision Systems (ICVS), 2017, pp. 195–204.

[2] M. Yang and G. Thung, “Classification of trash for recyclability status,” CS229 Course Project Report, Stanford University, 2016.

[3] Q. Zhang, Q. Yang, X. Zhang, Q. Bao, J. Su, and X. Liu, “Waste image classification based on transfer learning and convolutional neural network,” Waste Management, vol. 135, pp. 150–157, 2021.

[4] M. I. B. Ahmed, R. B. Alotaibi, R. A. Al-Qahtani, R. S. Al-Qahtani, S. S. Al-Hetela, K. A. Al-Matar, N. K. Al-Saqer, A. Rahman, L. Saraireh, M. Youldash, and G. Krishnasamy, “Deep learning approach to recyclable products classification: Towards sustainable waste management,” Sustainability, vol. 15, no. 14, p. 11138, 2023.

[5] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal speed and accuracy of object detection,” arXiv:2004.10934, 2020.

[6] W. Li, K. Liu, L. Zhang, and F. Cheng, “Object detection based on an adaptive attention mechanism,” Scientific Reports, vol. 10, p. 11307, 2020.

[7] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” in Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV), 2021, pp. 10012–10022.

[8] S. Mishra, R. Yaduvanshi, P. Rajpoot, S. Verma, A. K. Pandey, and D. Pandey, “An integrated deep-learning model for smart waste classification,” Environmental Monitoring and Assessment, vol. 196, p. 279, 2024.

[9] Y. Huang, D. Jiao, X. Huang, T. Tang, and G. Gui, “A hybrid CNN transformer network for object detection in optical remote sensing images: Integrating local and global feature fusion,” IEEE Trans. Geosci. Remote Sens., 2023.

[10] M. S. Sivakumar, T. Gurumekala, H. Rahul, N. Haldar, and H. Singh, “Design and development of smart internet of things–based solid waste management system using computer vision,” Environmental Science and Pollution Research, vol. 29, pp. 64871–64885, 2022.

[11] B. S. Devi, D. Gupta, and R. P. Singh, “Yolo-based deep learning techniques for identifying floating bottles in inland water: A comprehensive analysis,” in Proc. 15th Int. Conf. Computing Communication and Networking Technologies (ICCCNT), 2024, pp. 1–6.

[12] T. Ganguly, R. P. Singh, and P. Kumar, “Self-attention based ResNet model for cervical cancer detection,” in Proc. Int. Conf. Informatics (ICI), 2023, pp. 1–6.

[13] N. S. Rupak, N. Rayvanth, P. Kushank, and R. P. Singh, “An investigation into YOLO-v8 model optimization for small object detection in UAVs using attention mechanism,” in Proc. 15th Int. Conf. Computing Communication and Networking Technologies (ICCCNT), 2024, pp. 1–7.

[14] B. S. Devi, K. V. Nagaraja, and R. P. Singh, “Optimization enhancements for faster R-CNN in floating bottle detection,” in Proc. IEEE 11th Uttar Pradesh Section Int. Conf. Electrical, Electronics and Computer Engineering (UPCON), 2024, pp. 1–7.

Authorized licensed use limited to: Amrita School of Engineering. Downloaded on August 24,2026 at 17:11:52 UTC from IEEE Xplore. Restrictions apply.